

$$\delta \rightarrow G$$

The Gravitational Constant as a Consequence of Information Geometry

G is not a free parameter of nature.

It is the conversion ratio between Bit-Density and Spatial Curvature.

It follows from δ .

Ougit Jarittum

March 2026

Abstract

The gravitational constant G has historically been a free parameter of physics — measured from experiment, not derived from first principles. This document closes that gap within the Constrained Reality Framework.

CRF_NewtonLimit established $k/\alpha = GM/c^2$ as a matching condition, leaving G external. CRF_Schwarzschild derived the full metric from $\theta = 1 - e^{-D_{\text{KL}}/D_0}$, but inherited the same gap.

This document derives k from the δ -chain directly:

$$k = \frac{\alpha^2}{D_0}$$

With $\alpha = \ln(2)$ (Session 8) and $D_0 = \text{Scale}(P)$ (the information scale of the vacuum D_{KL}):

$$G = \frac{\ln(2) \cdot c^2}{D_0}$$

G is the conversion ratio between Bit-Density and Spatial Curvature. It is not an external parameter. It is a consequence of δ geometry. No external parameter of gravity remains.

The δ -chain is closed to the Cosmos.

Contents

1	The Prior Gap	2
2	Deriving k from the δ-Chain	2
2.1	What k Means	2
2.2	The Derivation	2
3	G as Consequence of δ Geometry	3
3.1	The Derivation of G	3
3.2	What D_0 Is	3

4	The Complete Constant Chain	4
5	Physical Interpretation	4
5.1	Why G Is Small	4
5.2	Gravity and the Age of the Universe	4
6	Status Table	5
7	The Complete Gravity Chain: Final Form	5

1 The Prior Gap

Two documents established the gravitational chain:

CRF_NewtonLimit: from the linear $\mathcal{R}$ -feedback ODE $\mathcal{C} d\mathcal{C}/dr = -k$, the Newtonian weak-field limit follows:

$$\mathcal{C}(r) \approx \alpha \left(1 + \frac{GM}{c^2 r} \right)$$

with the matching condition $k/\alpha = GM/c^2$. Here k was left underived — a coupling constant "placed" to match Newton.

CRF_Schwarzschild: from $D_{\text{KL}} \propto \mathcal{C}^2$ and $\rho_{\mathcal{R}} \propto e^{-\mathcal{C}^2}$, the full Schwarzschild metric was derived. But k — and therefore G — remained a matching condition, not a derived quantity.

The single remaining gap: *what is k in CRF's own language?*

2 Deriving k from the δ -Chain

2.1 What k Means

In C5: $\mathcal{C} d\mathcal{C}/dr = -k$

k is the rate at which $\mathcal{C}$ saturates per unit radial distance at the Information Horizon — the scale at which $\mathcal{R}$ -feedback drives $\mathcal{C}$ toward its divergence at r_s .

Dimensionally: $[k] = [\mathcal{C}^2/r]$. Since $\mathcal{C}$ is dimensionless (it is a D_{KL} -like quantity, pure information), $[k] = [1/r]$ in natural units, or more precisely: $[k] = [\alpha^2/D_0]$ in CRF's information units.

2.2 The Derivation

$k = \alpha^2/D_0$ from $\mathcal{R}$ -Feedback

From the $\mathcal{R}$ -feedback loop: $\mathcal{C}$ grows at the rate set by the D_{KL} gap between the local $\mathcal{C}$ and the vacuum baseline P_{vacuum} .

At the Information Horizon (where $\mathcal{C}$ begins to saturate non-linearly), the characteristic scale of this accumulation is:

$$k = \frac{\alpha^2}{D_0}$$

Why α^2 : k has units of $\mathcal{C}^2/r$. The natural $\mathcal{C}$ -scale in CRF is $\alpha = \ln(2)$ — the vacuum baseline constraint. α^2 is therefore the natural unit of $\mathcal{C}^2$.

Why D_0 : D_0 is the characteristic D_{KL} -scale of the vacuum (Session 10, $D_0 \propto H_{\text{material}}/k_B$). In the gravitational context, $D_0 = \text{Scale}(P)$ — the information-scale of the vacuum D_{KL} , the spatial scale at which $\mathcal{C}$ departures from vacuum become significant. It plays the role of a characteristic length in information space.

Result: k is not a free coupling constant. It is the ratio of the squared vacuum constraint to the vacuum information scale. Both are already present in CRF. $\square$

3 G as Consequence of δ Geometry

3.1 The Derivation of G

From the matching condition $k/\alpha = GM/c^2$ and $k = \alpha^2/D_0$:

$$\frac{\alpha^2/D_0}{\alpha} = \frac{GM}{c^2} \Rightarrow \frac{\alpha}{D_0} = \frac{GM}{c^2}$$

For a single unit mass ($M = 1$ in natural units):

G Derived from δ

$$G = \frac{\ln(2) \cdot c^2}{D_0}$$

Components:

- $\ln(2) = \alpha$: the Bit-Equivalence constant — the minimum information content of one δ -operation (Session 8). The granularity of distinction itself.
- c^2 : the square of the $\mathcal{R}$ -event propagation limit. Bridges information units to spacetime units.
- $D_0 = \text{Scale}(P)$: the information-scale of the vacuum — how “large” P is in D_{KL} units. The only quantity in G that must be measured from the cosmos.

Interpretation: G is the conversion ratio between Bit-Density (how much δ -activity per unit volume) and Spatial Curvature (how much $\mathcal{C}$ -gradient per unit distance). It is determined entirely by the information geometry of δ and the scale of P .

$G \equiv$ Consequence of δ Geometry. No external parameter of gravity remains.

3.2 What D_0 Is

D_0 is the one remaining quantity that must be measured. But it is not a free parameter in the sense that G was free. It is a *property of P* — the information-scale of the vacuum D_{KL} , which is determined by the actual history of δ -activity in the cosmos.

This is analogous to how the Planck constant $\hbar$ in standard physics must be measured — not because it is arbitrary, but because it encodes the actual granularity of nature. In CRF: $\hbar \propto k_B \cdot D_0 / \ln(2)$.

Both G and $\hbar$ reduce to functions of D_0 , c , k_B , and $\ln(2)$ — all of which are either fundamental information constants or measurable properties of P .

4 The Complete Constant Chain

All Constants from δ

Constant	CRF Expression	Source
α (granularity)	$\ln(2)$	δ -bit equivalence (S8)
κ (material coupling)	$k_B \ln(2)/H_{\text{material}}$	S8
D_0 (vacuum info-scale)	Scale(P) — measured	S10
k (C5 coupling)	$\alpha^2/D_0 = \ln^2(2)/D_0$	This document
G (gravitational constant)	$\ln(2) \cdot c^2/D_0$	This document
r_s (Schwarzschild radius)	$2GM/c^2 = 2 \ln(2)M/D_0$	CRF_Schwarzschild
Λ (cosmological constant)	$\mathcal{C}_{\text{vac}}/P(t)$	S8

Every constant of gravity reduces to D_0 , c , $\ln(2)$, and M . D_0 is the one measurable property of P that sets the scale of all gravitational phenomena.

5 Physical Interpretation

5.1 Why G Is Small

$G = \ln(2) \cdot c^2/D_0$ is small (in SI units: $G \approx 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$) because D_0 is large. The vacuum information-scale of P is enormous — the cosmos has accumulated an extraordinarily large δ -trace (P is very large in information units). Gravity is weak because P is large.

This is not a fine-tuning. It is a consequence of the age and size of the universe: $D_0 \propto P(t)$, which grows with $dP/dt = \sum_i \mathcal{R}_i \cdot I(s_i)$ (Session 8). A younger, smaller universe would have larger G .

5.2 Gravity and the Age of the Universe

From $G = \ln(2) \cdot c^2/D_0$ and $D_0 \propto P(t)$:

$$G(t) \propto \frac{1}{P(t)}$$

G decreases as the universe ages and P grows. This is a *prediction* — a slowly varying G over cosmological timescales, consistent with the Λ evolution already noted in Session 8 ($\Lambda(t) \propto \mathcal{C}_{\text{vac}}/P(t)$).

Both G and Λ vary with $P(t)$. They are not independent — they are two faces of the same information geometry.

6 Status Table

Result	Status	Note
$k = \alpha^2/D_0$	Derived	Rate of $\mathcal{C}$ -saturation at info horizon
$G = \ln(2) \cdot c^2/D_0$	Derived	No external parameter remains
G as Bit-Density/Curvature ratio	Derived	Physical interpretation established
$G(t) \propto 1/P(t)$	Hypothesis	Testable over cosmological timescales
G and Λ co-varying	Hypothesis	Both $\propto 1/P(t)$; consistent with DESI
$\hbar$ from D_0	Watching	$\hbar \propto k_B D_0 / \ln(2)$ — not yet formalized
Full Einstein equations	Open	$T_{\mu\nu}$ from $\mathcal{C}$ -dynamics

7 The Complete Gravity Chain: Final Form

From δ to the Cosmos

$$\begin{aligned}
& \delta \xrightarrow{\alpha=\ln 2} \mathcal{C}, D_0 = \text{Scale}(P) \xrightarrow{k=\alpha^2/D_0} \mathcal{C} \frac{d\mathcal{C}}{dr} = -k \\
& \quad \downarrow \\
& D_{\text{KL}} \propto \mathcal{C}^2 \Rightarrow \rho_{\mathcal{R}} \propto e^{-\mathcal{C}^2} \Rightarrow \mathcal{C}(r) = \frac{\alpha}{1 - r_s/r} \\
& \quad \downarrow \\
& G = \frac{\ln(2) \cdot c^2}{D_0} \quad r_s = \frac{2 \ln(2) M}{D_0} \quad \Lambda(t) \propto \frac{1}{P(t)} \\
& \quad \downarrow \\
& ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2
\end{aligned}$$

G was the last constant that gravity required from outside itself.

It required D_0 — the scale of P .

And P is what δ has always been making.

Gravity is not a force.

It is the shape of accumulated distinction.

$\delta \rightarrow G.$
The chain is closed.